
Toscanini 01 executive technology brief

invbench-p03-phe-biofouling156ae5155c73 · run r401-bench-p03-phe-biofouling · maturity
EARLY_TECHNICAL_EVALUATION · 3D class CONCEPTUAL_3D

Technology

inv:bench-p03-phe-biofouling:156ae5155c73

Bottleneck

- Problem Id: bench-p03-phe-biofouling
- Device: stainless steel plate heat exchanger in a district cooling loop
- Failure: biofilm and scale accumulation on plate channels reduces overall heat transfer coefficient and raises pressure drop between cleanings
- Failure Mode: fouling
- Constraint: sust...

Causal mechanism

- Mechanism: Real-time distributed fiber optic sensing combined with microcapsule-based self-healing agents that release when corrosion is detected
- Intervention: Embedding microstructured optical fiber sensors and corrosion-activated microcapsules containing healing agents throughout the stainless...

Differentiating architecture

- Collision Novelty Risk: UNRESOLVED_INSUFFICIENT_EVIDENCE
- Prior Art Status: UNRESOLVED_INSUFFICIENT_EVIDENCE
- Multi Source Directions: • Discovery: (sources_recorded, sources_errored)
- Destruction: (sources_recorded, sources_errored)
- Transfer: (sources_recorded, sources_errored)
- Reality: (s...

Expected technical advantage

UNKNOWN (not recorded)

Strongest supporting evidence

Evidence classification counts from the run's own verification: direct support: 10; partial support: 5; background: 0; analogy: 0; contradictory: 0; irrelevant: 1; direct-support spans are itemized in 04_EVIDENCE_SUMMARY.

Biggest uncertainty

FM_WITHOUT_QUANTITY_MATCHED_VERIFICATION:FM-DOM-005

Engineering status

EARLY_TECHNICAL_EVALUATION; 11 design inputs, 6 failure modes, 3 governing equations recorded

Decisive experiment

physical bench/clinical experiment — NOT simulation

Buyer decision

evaluate this package against the nine questions in 03_BUYER_DECISION_CARD; the cheapest falsification is named there with its acceptance rule

Epistemic boundary: DESIGNED / SIMULATED / EVIDENCE-SUPPORTED / EXPERIMENTALLY VERIFIED states come only from the canonical invention object. EXPERIMENTALLY VERIFIED requires real reality-loop evidence. Potential IP territory — formal legal review required. Toscanini performs technology discovery; patentability, claim drafting, infringement/FTO and legal opinions are matters for qualified IP counsel.